

Mark Alvin Cadangin

Full-Stack Software Engineer | Systems & Embedded Developer | DOST-SEI Scholar

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PROFESSIONAL SUMMARY

3rd-year BSIT student specializing in Software Development Technologies at West Visayas State University and a DOST-SEI Scholar (Batch 2024). Builds practical full-stack web applications and IoT edge systems using Java/Spring Boot, Next.js, Laravel, FastAPI, and embedded C++. Focused on relational database integrity, local fail-safe firmware logic, and reliable offline operations.

EDUCATION

West Visayas State University
Bachelor of Science in Information Technology -- Major in Software Development Technologies
• DOST-SEI Scholar (Batch 2024), Department of Science and Technology -- Science Education Institute
• CICT Academic Excellence: Gold Medalist (1st Year) | Silver Medalist (2nd Year)

Iloilo City, Philippines
2024 - Present

TECHNICAL SKILLS

Languages: Java, Kotlin, Python 3.12, C++, TypeScript, JavaScript, PHP, SQL (PostgreSQL, MySQL, SQLite)
Backend & APIs: Spring Boot 3.5, FastAPI, Laravel 13, RESTful APIs, Asynchronous Background Tasks
Frontend & Mobile: Next.js 15/16, React 19, Jetpack Compose (Android), Tailwind CSS, Vite, HTML5, CSS3
Databases & DevOps: PostgreSQL, MySQL 8.4 Primary-Replica, SQLite WAL, Firebase RTDB, Docker Compose, Git, Linux / Bash
Hardware & IoT: ESP32, ESP8266, RS-485 Differential Bus, Modbus Protocol, BLE, Ultrasonic Sensors, CJX2 AC Contactors

ENGINEERING PROJECTS

SmartFlow -- Field-Deployed IoT Water-Pump Controller
C++, ESP32, ESP8266, RS-485 Differential Bus, CRC16, Kotlin, Jetpack Compose, Firebase RTDB | Leon, Iloilo
• Built and field-installed a two-node controller over a 40m CAT6 link, eliminating motor electrical noise with RS-485 and CRC16.
• Wrote C++ firmware fail-safes to protect the 1.5 HP motor: 15s dry-run cutoff, 45-min runtime limit, and CJX2 contactor control.
• Programmed all shutdown decisions to execute locally on the ESP32, ensuring the pump halts safely even when Wi-Fi drops.
• Developed a native Android app in Kotlin and Jetpack Compose for BLE provisioning, live water level monitoring, and telemetry.

2026 | Personal Project

CONVERA -- Evidence-Driven Requirements Validation Platform
Next.js 15.2, FastAPI, Python 3.12, SQLite WAL (38 tables), TypeScript, Tailwind CSS v4, Notion/Zotero API
• Built a validation platform that structures raw research notes into formal software proposals following IEEE 830 standards.
• Created an interactive evidence graph linking literature claims to software requirements across 38 SQLite tables with 338 tests.
• Implemented async validation services in FastAPI and Python to compute viability scores using explicit mathematical formulas.
• Built direct export integrations for Notion, Zotero (BibTeX/CSL-JSON), and GitHub Issues with SHA-256 hashed audit records.

2026 - Present | Personal Project

HavenStay BHMS -- Bed-Level Property Management Platform
Next.js 16, React 19, Laravel 13, MySQL 8.4 Primary-Replica, Docker Compose, Tailwind CSS
• Built a bed-level rental system tracking room vacancies, tenant deposits, and prorated calculations for shared water and power meters.
• Wrote 45 database triggers in MySQL 8.4 to record before-and-after change snapshots on tenant balances directly at the DB level.
• Configured MySQL 8.4 primary-replica replication in Docker Compose to separate monthly billing calculations from live booking writes.
• Containerized the full stack with Docker Compose, automating database migrations and multi-container local testing environments.

2026 | Course Project

Faith Laundry Shop -- Laundry Order Management System
Next.js 15, Spring Boot 3.5, Java 21, PostgreSQL, Inno Setup Windows Installer
• Built an order management system modeled on an active shop workflow, replacing paper notebooks with digital intake & tracking.
• Implemented dynamic pricing with price-snapshotting in Spring Boot, locking transaction rates against future catalog price changes.
• Developed a zero-login customer portal allowing clients to check live wash, dry, and fold status by scanning receipt QR codes.
• Packaged an offline-first Windows installer with Inno Setup provisioning PostgreSQL and running the backend as a background service.

2026 | Course Project